



Lab 09 – Worksheet

Name: Faaz Shahab, Mohib Ul Mehdi, Basim Hussain

ID: fs09911,
mm09730,
mh09732

Section: T2

Task 1

Main Control TT:

Instruction	Opcode	RegWrite	ALUSrc	MemRead	MemWrite	MemtoReg	Branch	ALUOp
R-type	0110011	1	0	0	0	0	0	10
ADDI	0010011	1	1	0	0	0	0	10
Load (lw/lwlb)	0000011	1	1	1	0	1	0	00
Store (sw/sb/ra)	0100011	0	1	0	1	X	0	00
BEQ	1100011	0	0	0	0	X	1	01

ALU Control TT:

ALUOp	Funct 3	Funct 7	Operation
00	X	X	ADD
01	X	X	SUB
10	000	0000000	ADD
10	000	0100000	SUB
10	111	0000000	AND
10	110	0000000	OR
10	100	0000000	XOR
10	001	0000000	SLL
10	101	0000000	SRL



Task 2

Required Verilog Files attached in the folder uploaded in Github

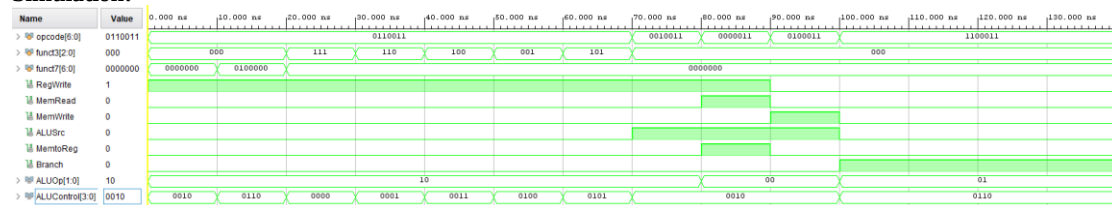
Synthesis proof:

✓ Synthesis

✓ Synth Design (synth_design)

synth_1_synth_report_utilization_0	report_utilization	3/17/26, 10:44 AM	6.7 KB
synth_1_synth_synthesis_report_0		3/17/26, 10:44 AM	11.9 KB

Simulation:



Task 3

Required Verilog Files attached in the folder uploaded in Github

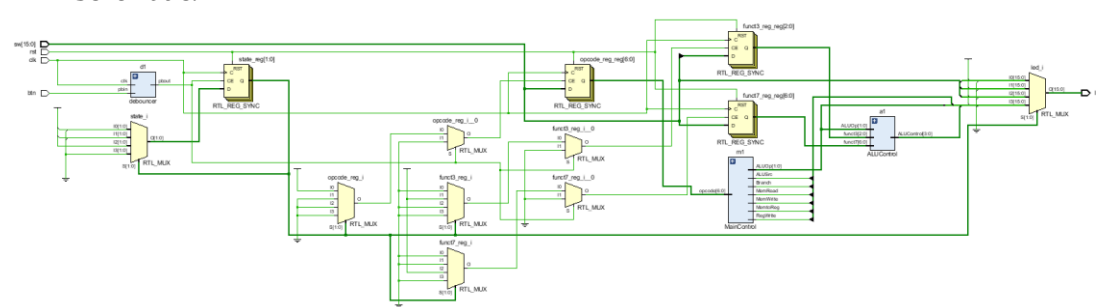
Synthesis proof:

✓ Synthesis

✓ Synth Design (synth_design)

synth_1_synth_report_utilization_0	report_utilization	3/17/26, 11:09 AM	7.0 KB
synth_1_synth_synthesis_report_0		3/17/26, 11:09 AM	18.1 KB

RTL Schematic:





Assessment Rubric

Lab 09: Implementing Nested Procedures and Sorting

	Task No.	LR 2 Code	LR 5 Results	LR 11 (Design)
In-Lab	Pre-Lab	-	/15	-
	Task 1	-	-	/10
	Task 2	/20	/20	-
	Task 3	-	/15	-
Total Points		/20	/50	/10
CLO Mapped		CLO 1		

Affective Domain Rubric		Points	CLO Mapped
AR7	Report Submission & Git Upload	/10 & /10	CLO 1

CLO	Total Points	Points Obtained
1	100	
Total	100	

For description of different levels of the mapped rubrics, please refer to the Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics provided here.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/Simulation Model/Network Model	Program/code/simulation model/network model doesnot implement the requiredfunctionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and doesnot produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full commandon the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorlyconstructed with erroneousresults. Titles, captions, units are not mentioned. Data is presented in anobscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are notaccurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minorerrors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captionsand proper units are mentioned. Data presentation is systematic.
AR9	Report Content/Code Comments	No summary provided. The number/amount of tasks completed below the level of satisfaction and/or submitted late	Couldn't provide good summary of in-lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots andfigures provided	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	Outstanding Summary of In-Lab tasks. All task completed and explained well, submitted on time, good presentation of plotsand figure with proper label, titles and captions